

‘Help, my teacher is pressuring me!’

The Role of Students’ Coping with Controlling Teaching in Motivation and Engagement

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Abstract

Ample research has shown that a controlling teaching style is detrimental for students' motivation and engagement in school. However, little is known about how students cope with such a teaching style. Therefore, grounded in Self-Determination Theory, the present study examines the role of four coping strategies (i.e. oppositional defiance, compulsive compliance, negotiation and accommodation) in students' motivation and engagement. A repeated measures design with four weekly assessments was used to examine associations both at the between- and within-student level. The sample consisted of students from 4th to 6th grade ($N = 433$; 51% boys, $M_{age} = 10.6$ years). The findings showed that oppositional defiance exacerbated associations between controlling teaching and amotivation. The moderating role of both compulsive compliance and negotiation was limited. Accommodation buffered associations between controlling teaching and amotivation and low engagement at the within-person level of analysis. In addition to a moderating role, we also found evidence for a mediating role of coping. Specifically, controlling teaching related positively to oppositional defiance and compulsive compliance, which, in turn, were related to maladaptive student outcomes. Moreover, controlling teaching related positively to negotiation, which, in turn, related positively to adaptive student outcomes. Overall, the results underscore students' active role in teacher-student dynamics. Limitations and directions for future research are discussed.

According to Self-Determination Theory (SDT; Ryan & Deci, 2017), students' quality of motivation for school is an important predictor of students' engagement and well-being at school (Bundick, et al., 2014; Guay et al., 2008; Ryan & Deci, 2000). Indeed, abundant research has shown that high quality motivation (i.e., autonomous motivation) is positively related to behavioral (e.g., engagement and achievement; Fredricks et al., 2004; Guay & Bureau, 2018; Vansteenkiste et al., 2004), affective (e.g., well-being; Deci & Ryan, 2008; Levesque et al., 2004), and cognitive (e.g., depth of learning; Vansteenkiste et al., 2004) outcomes. A low quality of motivation (such as controlled motivation and amotivation) instead is related to negative behavioral (e.g., procrastination; Mouratidis et al. 2018; Senécal et al., 2010), affective (e.g., boredom; Ntoumanis., 2001; Skinner et al., 2008), and cognitive (e.g., superficial cognitive processing; Vansteenkiste et al., 2005) outcomes. Unfortunately, educational research has shown that students' autonomous motivation declines, while controlled motivation and disengagement increase throughout students' school career and in particular in the transition from elementary to secondary education (Gillet et al., 2011; Skinner, et al., 2008). One factor playing an important role in students' motivation and engagement is whether teachers adopt a motivating versus a more demotivating teaching style (Reeve, 2009).

Controlling Teaching

Recently, a small but growing body of research based on SDT has begun to examine the dark side of teaching, thereby focusing on controlling teaching (Haerens et al., 2016). A controlling teaching style is characteristic of teachers who ignore the students' perspective and who pressure students to think, feel, or act in a particular way (Reeve, 2009). Controlling teachers engage in various pressuring behaviors aimed at enforcing student compliance, including externally controlling tactics, such as threatening with punishment and controlling

language, and more internally controlling tactics, such as shaming and guilt induction (Assor et al., 2005; Soenens et al., 2012). A controlling teaching style is assumed to undermine students' motivation, engagement, and achievement (Reeve, 2009). Indeed, a growing number of studies showed that controlling teaching is positively related to unfavorable student outcomes, such as controlled motivation (Assor et al., 2005; De Meyer et al., 2014), amotivation (Assor et al., 2005; De Meyer et al., 2014), disengagement (Van den Berghe et al. 2016), and ill-being (Assor et al., 2005; Liu et al. 2017). At the same time, controlling teaching relates negatively to positive student outcomes, such as autonomous motivation (Soenens et al., 2012), engagement (Assor et al., 2005), achievement (Soenens et al., 2012), and deep level learning (Vansteenkiste et al., 2005).

Most studies on controlling teaching thus far have been cross-sectional in nature. As such, these studies address the role of controlling teaching in interindividual motivational differences between students, without providing information about how controlling teaching relates to intra-individual motivational differences within students across time. Only few studies have examined within-person fluctuations across time in perceived controlling teaching style and students' motivation. Bartholomew et al. (2018) for instance conducted a longitudinal study with middle school students that consisted of three measurements across the school year. The results showed that on occasions when students perceived **higher levels of** teachers' use of control **(relative to their average perception)**, they simultaneously reported **more** amotivation and **less** autonomous motivation **(compared to their individual average)**. Similarly, Patall et al. (2018) demonstrated, in a diary study among 208 high school students, that daily perceptions of demotivating teacher practices (including controlling practices) predicted daily controlled motivation and disengagement.

According to SDT, controlling teaching undermines students' motivation and engagement because such a teaching style thwarts children's basic psychological needs, particularly the need for autonomy (Reeve et al., 2004). The need for autonomy reflects the need to experience a sense of volition and psychological freedom. When the need for autonomy is satisfied, students experience freedom to behave in a way that is concordant with their personal interests, preferences, and values. As such, they feel authentic in their behavior, which is assumed to be essential for students' motivation, engagement, and well-being in school (Deci & Ryan, 2000; Niemiec & Ryan, 2009; Soenens et al., 2018). Not only is controlling teaching assumed to hinder satisfaction of the need for autonomy, it would also actively thwart this need, resulting in autonomy frustration (Vansteenkiste & Ryan, 2013). While autonomy satisfaction involves feelings of volition and self-endorsement, autonomy frustration would manifest through feelings of pressure and obligation, with such feelings increasing the risk for maladaptive student outcomes.

Both cross-sectional (e.g., Haerens et al., 2015) and longitudinal studies (e.g., Bartholomew et al., 2018) confirmed the intervening role of autonomy frustration in **associations among** perceived controlling teaching and students' maladaptive outcomes. Based upon these findings, one could argue that classroom interventions should focus on decreasing controlling teaching and on promoting autonomy-supportive teaching as an alternative. Indeed, studies showed that teachers can be trained to adopt a more autonomy-supportive style, instead of a controlling style (Aelterman et al., 2014), with ensuing benefits for students' motivation and engagement (Cheon et al., 2012; Cheon & Reeve, 2013). However, these studies also show that effect sizes of these interventions are in the small to medium range. Additionally, teachers that are highly controlling may be less inclined to participate in autonomy-supportive trainings

(Cheon & Reeve, 2015). Therefore, it is important to examine students' agentic role in controlling teaching by considering students' coping strategies. Ultimately, these coping strategies may represent additional targets for intervention that may complement teaching interventions.

Students' Agency in Coping with Controlling teaching

All too often, students are considered as passive recipients of teachers' instructions and practices in the classroom (Reeve, 2013). This is unfortunate, because both theory (e.g., classroom ecology paradigm; Doyle, 1977; Hastie & Siedentop, 2006), and empirical research (e.g., Jang et al., 2012; Sarrazo et al., 2006; Van den Berghe et al., 2016) suggest that students' behavior plays a significant role in student-teacher dynamics. By addressing students' role in the interaction with teachers as well, a more complete picture of teacher-student dynamics can be drawn. Also SDT supports the idea of children displaying high levels of agency in interacting with socialization figures (Reeve, 2013; Reeve & Tseng, 2011; Soenens & Vansteenkiste, 2020). Instead of being passive recipients of a controlling environment, and being affected by it as such, students would take matters into their own hands by actively trying to handle the frustration of their psychological need for autonomy (Legault et al., 2017; Ryan et al., 2019; Sheldon, 2011). However, pupils' active and agentic role in these detrimental teacher-student dynamics is much less explored. One important aspect of students' agency is the way students cope with controlling teacher behavior. Specifically, students are expected to differ in the way they respond to threats to their psychological needs. An examination of students' coping **with** controlling teaching advances the literature on a fundamental and theoretical level. Indeed, rather than assuming that adolescents are affected quite passively and uniformly by a controlling teacher, it is important to examine whether and how students can actively influence how they are impacted by a controlling

teacher. As such, this research highlights students' agency in the classroom, a theme that receives insufficient attention in the educational literature (Reeve & Tseng, 2011). Previous research did attend to one aspect of students' agency, that is, their appraisals of teacher behaviors. This research has shown that students differ in their appraisals of teachers' behaviors, with some students being more sensitive to controlling behavior (De Meyer et al., 2016; Zhou et al., 2012). Students' characteristics determine to some extent the degree to which a potentially controlling teaching practice is perceived as controlling (Vansteenkiste et al., 2020). An important next step in this literature is to examine whether students also differ in their coping responses following appraisals of controlling teaching. **Students' coping results from the appraisal or meaning students assign to the teacher's practices, with both unconscious and conscious processes playing a role in students' appraisals.** A complete understanding of students' agency in the teacher-student relationship indeed requires attention to both students' appraisals and their coping responses (Soenens & Vansteenkiste, 2020).

Herein, we rely on a model of coping developed by Skinner and colleagues (Skinner & Edge, 2002; Skinner et al., 2003). Skinner and colleagues reviewed 400 ways of coping and classified them into twelve families of coping. Each of these families are organized around SDT's three basic psychological needs. In the present study, we focus on coping with autonomy frustration, because controlling teaching primarily thwarts the need for autonomy (Reeve et al., 2004). This model distinguishes between two dimensions, that is, (a) whether students concede to or defend against the controlling teacher and (b) whether students respond in an autonomous or self-determined way (i.e., staying true to themselves) or whether they respond in a non-autonomous or reactive way (i.e., giving up on personal priorities). By crossing those two criteria, four coping responses can be identified (see Figure 1).

A first coping strategy is oppositional defiance. This response refers to students' blunt rejection of the demands made by the teacher and their tendency to do the opposite of what the teacher requested, thereby completely disregarding the teacher's authority (Aelterman et al., 2016; De Meyer et al., 2016a; Skinner & Edge, 2002; Van Petegem et al., 2015). Oppositional defiance is considered a non-autonomous form of defense, because the resistance is completely driven by opposing the teacher-induced pressure and does not result from internal goals or values. Second, compulsive compliance involves rigid conformity to the teacher's pressure and anxious obedience to the teacher's demands. Because students suppress their own preferences to comply with the teacher's demands, the concession is non-volitional. Hence, compulsive compliance is also considered a non-autonomous coping response.

In addition to these two non-autonomous, presumably rather dysfunctional coping styles, two more autonomous and potentially more adaptive responses are identified. Negotiation involves engaging in a constructive and flexible dialogue with the teacher, thereby trying to find a consensus between the student's own interests and preferences and the teacher's demands. Negotiation is considered a more autonomous form of coping, because students prioritize personal goals and preferences in the search for a compromise with the teacher (Parkin & Kuczynski, 2012). A final coping response is accommodation. This response implies a flexible adjustment of the pupils' own goals and preferences, which can be done by cognitively restructuring the request, by accepting the constraining situation, or by focusing on more important goals (Brandtstädter & Rothermund, 2002; Compas et al. 2001; Morling & Evered, 2006). Because the student understands the added value of the demand and thus volitionally accepts the teacher's demand, this concession is also considered to be autonomous in nature.

Although these coping responses have been discerned conceptually, only few studies have examined these coping responses empirically. The limited empirical work available was conducted in the context of parenting. These studies showed that controlling parenting evokes more non-autonomous coping responses (i.e., oppositional defiance or compulsive compliance) (Brenning et al., 2019; Chen et al., 2016; Van Petegem et al., 2015; 2017, 2019; Vansteenkiste et al., 2014) and less autonomous coping responses (i.e., negotiation or accommodation) (Brenning et al., 2019; Chen et al., 2016; Van Petegem et al., 2017).

The few relevant studies in the educational context have focused only on oppositional defiance as a coping response to controlling teaching. Haerens et al. (2015) showed that perceived controlling teaching was positively related to oppositional defiance. This finding was confirmed in an experimental study where students watched video-based vignettes of an autonomy-supportive versus controlling teacher (De Meyer et al., 2016a). In turn, studies have shown that oppositional defiance related negatively to need satisfaction, autonomous motivation, and engagement, and positively to need frustration, amotivation, resentment towards the teacher, and a greater use of more antisocial strategies such as acting out, cheating, and truancy (Aelterman et al., 2018; Aelterman et al., 2016; De Meyer et al., 2016a). Although these studies have begun to examine the association between controlling teaching, oppositional defiance, and student outcomes, the precise role of oppositional defiance in effects of controlling teaching has not been determined. Moreover, to the best of our knowledge, no research to date considered the role of compulsive compliance, negotiation, and accommodation as ways of coping with a controlling teaching style.

Conceptually, coping responses can play two different roles in the associations among controlling teaching and student outcomes. First, coping responses could affect the degree to

which students suffer from a controlling environment (in the case of dysfunctional coping responses) or display resilience against controlling teaching (in the case of constructive coping responses) (i.e., moderation). A second possibility is that coping responses could explain at least partly the detrimental effects of a controlling environment (i.e., mediation). The more students are exposed to a controlling classroom environment, the more they may engage in dysfunctional (rather than more constructive) responses, with these responses in turn relating to motivational, emotional, and behavioral problems. The question whether coping responses primarily play a moderating role or a mediating role has been a topic of debate for decades. Whereas some theorists see coping as a mediating mechanism between stress and outcomes (e.g., Lazarus & Folkman, 1984), others proposed that coping should be examined as a moderating variable (e.g., Aldwin, 1994; Cohen & Wills, 1985). Currently, scholars no longer assume that the moderating and mediating roles of coping are mutually exclusive. Instead, it is increasingly argued that it is important to consider both possibilities simultaneously (e.g., Dardas & Ahmad, 2015; Jose & Huntsinger, 2005). Moreover, **up to date**, neither the possibility of a moderating role of coping responses, nor the possibility of a mediating role of coping responses in the association between a controlling teaching style and students' outcomes has been examined in the educational context. Therefore, in the current study, we examine both the moderating and mediating role of coping in associations **among** controlling teaching and students' functioning.

The Present Study

Although both theory and research suggest that teacher-student interactions are highly dynamic in nature, all too often, students are considered to be passive recipients in the classroom (Reeve, 2013). A growing body of research, based on SDT, found that a controlling teaching style has detrimental effects on students' quality of motivation, engagement, and well-being

(Haerens et al., 2016). However, there is a dearth of studies examining how students cope with controlling teaching and the associated threat to their need for autonomy. This study aims to add to the literature by considering the possible moderating as well as mediating role of four different coping responses (i.e., oppositional defiance, compulsive compliance, negotiation, and accommodation) in associations among perceived controlling teaching and students' motivation and engagement. Against the background of our overall aim to examine the role of students' coping, we formulate the following hypotheses. First, regarding the moderating role of coping, we predict that oppositional defiance and compulsive compliance will exacerbate the associations among controlling teaching and aspects of students' problematic functioning (i.e., low autonomous motivation, controlled motivation, amotivation, and low engagement), whereas negotiation and accommodation would buffer these associations. Second, regarding the mediating role of coping, in line with the distinction between a bright and a dark pathway in SDT (Haerens et al., 2015), we hypothesize that controlling teaching will positively relate to oppositional defiance and compulsive compliance, which in turn, will positively relate to negative student outcomes (controlled motivation and amotivation). We expect that controlling teaching will be unrelated or negatively related to negotiation and accommodation, with these coping strategies in turn being positively related to positive student outcomes (autonomous motivation and engagement). Methodologically, this study adopts a short-term repeated measures design with 4 weekly assessments, as to examine the role of coping responses both at the level of between-student differences and at the level of within-student fluctuations in exposure to controlling teaching. The level of between-student differences addresses the question whether students who on average (i.e., across time) engage more often in maladaptive (or adaptive) coping strategies in response to controlling teaching fare worse (or better) in terms of their

quality of motivation and engagement compared to other students. The level of within-student fluctuations refers to variability in these processes across time. In other words, an analysis at the within-student level allows to examine whether students who, in a given week, respond in a more maladaptive (or adaptive) way to controlling teaching, compared to their own average coping response, simultaneously fare worse (or better) in terms of motivation and engagement in that week, compared to their own average level of motivation and engagement. Coping in particular needs to be examined both at the between- and within-person level of functioning (Curran & Bauer, 2011; Hamaker et al., 2015; Keijsers, 2016). A child's engagement in more effective coping is thought to influence the child's intrapersonal functioning. So, in addition to the question whether there are differences between students due to their coping, it is important to also consider if students function better in the weeks they use more constructive coping strategies, compared to weeks when they use more non-autonomous coping strategies. Such findings would be especially interesting from an applied perspective, because the within-person level is where effects of interventions and preventions operate primarily.

At the between-person level, we took into account associations between students' gender, grade level and family structure on the one hand and the study variables on the other hand. Doing so was deemed important because previous studies have shown that these background variables are related to students' motivation, engagement, and their coping (Martin, 2009; Skinner & Saxton, 2019; Zimmer-Gembeck and Skinner, 2011).

Method

Participants and Procedure

The sample consists of 433 students¹ from 23 classes of 8 randomly selected elementary schools in the Dutch speaking part of Belgium ($M_{age} = 10.6$ years, $SD = 0.98$, range = 9 to 13 years, 51.4% boys). In this sample, 29% of the children were in 4th grade, 28% were in 5th grade, and 43% were in 6th grade. The class size ranged from 10 to 30 students.

In April, the students were asked to fill out the questionnaire for four consecutive weeks. All questionnaires were administered during class hours under the guidance of the classroom teacher who was informed about the procedures by the researchers. Teachers were asked not to walk around in the classroom during data collection. To assure confidentiality, the participants returned the filled out questionnaires to the teacher in a sealed envelope. The teachers then returned the sealed envelopes to the researchers. Of the full sample, 4.2%, 4.2%, 6.9%, and 13.3% of the data were missing at week 1, week 2, week 3, and week 4, respectively. To estimate missing data, we relied on Full Information Maximum Likelihood (FIML; Wothke, 2000), which enables to use all available data points, even for students with missing data, thereby maximizing statistical power. The procedure used in this study was approved by the ethical committee of [details removed for peer review] (nr. 2016/111).

Measures

All measures were administered each week for four consecutive weeks. To ensure feasibility of the data gathering in the repeated measured design, we used shortened versions of validated questionnaires. All items were scored on a 5-point Likert scale, ranging from 1 (*totally disagree*) to 5 (*totally agree*). Items can be found in Table 1 in appendix. **To assess internal reliability, McDonald's omega (McDonald, 1999) was computed for each scale (averaged over the four weeks).**

¹ The sample size was determined based on Monte Carlo simulations in Mplus.

Perceived Controlling Teaching Style. Children completed a short form of a recently developed 12-item scale to assess their perceived controlling teaching (De Meyer et al., 2016b). This shortened scale consisted of 6 items (e.g., “This week, my teacher threatened with sanctions when I was not doing what (s)he told me to do”). **On average, omega reliability was equal to .82.**

Quality of Motivation. Children reported on their quality of motivation by completing a short form of the adapted version of the Academic Self-Regulation Scale (Ryan & Connell, 1989; Vansteenkiste et al., 2009). In total, the scale consisted of 9 items, assessing autonomous motivation (3 items; e.g., “This week, I was motivated to study because it was personally important to me”), controlled motivation (3 items; e.g., “This, week, I was motivated to study because I would feel guilty if I wouldn’t do so”), and amotivation (3 items; e.g., “This week, I did not see any reason why I would do my best in school”). **Omega’s reliability was on average .81, .66, and .80 for respectively autonomous motivation, controlled motivation, and amotivation.**

Engagement. Children completed 3 items tapping into their own behavioral participation in the classroom (Skinner et al., 2008; e.g., “This week in class, I worked as hard as I could”). **On average, omega reliability was .74.**

Coping Responses. Children reported each week on how they dealt with feelings of pressure and obligation induced by their teacher. First, they read following instructions: ‘The next questions concern moments when you felt that the teacher obliged you to do something. More specifically, moments when you felt pressured to do something against your will’. These instructions were followed by an item stem: ‘On moments this week when I felt my teacher obliged me to do something...’, which was in turn followed by 12 items tapping into the four coping responses: oppositional defiance, compulsive compliance, negotiation, and

accommodation. Oppositional defiance was assessed using 3 items (Vansteenkiste et al., 2014; e.g., “I did the opposite of what was expected”). Children completed 3 items to assess compulsive compliance (Baudat et al., 2016; Skinner & Zimmer-Gembeck, 2007; e.g., “I did what the teacher asked of me but against my will”). Negotiation was assessed through 3 items of the adapted version of the Negotiation subscale of the Child Coping Questionnaire (CCQ; Chen et al., 2016; Finnegan et al., 1998; Ojanen & Perry, 2007; e.g., “I explained to the teacher why I did not agree”). Finally, to assess accommodation, children completed 3 items based on items from the Secondary Control subscales of the Responses to Stress Questionnaire (Connor-Smith et al. 2000) and the Acceptance subscale of the Cognitive Emotion Regulation Questionnaire (Garnefski and Kraaij 2006; e.g., “I tried to see that the teacher means well”). **Omega’s reliability was on average .58, .63, .68 and .65 for respectively oppositional defiance, compulsive compliance, negotiation, and accommodation.**

Plan of Analysis

As a first step, to examine the psychometric properties of the weekly variables, a series of confirmatory factor analysis (CFA) was performed. This was done for each measurement wave, resulting in 4 CFAs in total. These models included controlling teaching, coping (i.e., oppositional defiance, compulsive compliance, negotiation, and accommodation) and the outcome variables (i.e., autonomous motivation, controlled motivation, amotivation, and engagement). Each model had an RMSEA $\leq .06$, an SRMR $\leq .07$ **and a CFI $\geq .88$** , providing evidence for a good model fit (Hu & Bentler, 1998). Moreover, all factors loadings were significant at $p < .01$.

Because each student participated multiple times and students were nested in classes, multilevel models were performed with three levels (i.e., weekly level, student level, class level). Multilevel models were used to examine the role of coping responses in the effects of a perceived

controlling teaching style on students' quality of motivation and engagement both at the between-student level (reflecting differences between students) and within-student level (reflecting weekly deviations from an individuals' mean level). To examine the proportion of variance at the three levels (weekly measurements, student level, and class level), intercept-only models were estimated for all study variables. As a result, two intraclass correlations (ICCs) can be calculated, one representing the correlation situated at the class level (ICC_{class} ; between-group level) and one representing the correlation situated at the student level (ICC_{id} between-person level). Table 1 shows both ICCs for all study variables. As can be seen in the table, there is substantial covariation at both the between-person and within-person level, for all study variables. However, because for autonomous motivation, controlled motivation, and amotivation, the ICC at the class level was below 0.05, which can result in convergence problems (Preacher et al., 2010), we decided to run only two-level analyses, ignoring the class level. In these analyses, students represent the upper level (reflecting the between-person level) and the repeated measurements represent the lowest level (reflecting the within-person level) for these outcomes. Regarding engagement, we found more than 5% variation at the class level, so, for this variable, we decided to control for the class level. School level was ignored because the small number ($n = 8$) would lead to unreliable estimates (Maas & Hox, 2005). Moreover, we found that, on average, the school level explained less than 3% of the variance in the data (range 0-8%). Random intercepts, providing a separate, personal intercept for every student, were used to capture the correlation over time within each student.

To examine the main hypotheses, two models were tested, with one model testing the mediating hypothesis and the other model testing the moderating hypothesis. In the interest of comparability between studies, we relied on the model indirect command in Mplus to test

mediation, which is a very common and highly recommended test of indirect effects. Both the moderation and the mediation model included perceived controlling teaching, the four coping responses (i.e., oppositional defiance, compulsive compliance, negotiation, and accommodation) and four outcome variables (i.e., autonomous motivation, controlled motivation, amotivation, and engagement). In addition to the tested directional paths, correlations were added among the coping responses and among the outcome variables. Because of the nested structure in the data, we relied on the multilevel structural equation (MSEM) approach which was shown to reduce bias both when estimating interaction effects (Preacher et al., 2016) and indirect effects (Preacher et al., 2010). All models were estimated in Mplus 8.3 (Muthén & Muthén, 1998-2017). Model fit was evaluated by means of the following indices: the Comparative Fit Index (CFI), the Root Squared Error Approximation (RMSEA) and the Standardized Root Means Square Residual (SRMR). Values larger than 0.90 for CFI, and smaller than .06 for RMSEA and smaller than .08 for SRMR indicate a good fit (Hu & Bentler, 1999).

Results

Descriptive Statistics and Preliminary Analyses

Descriptive statistics (means and standard deviations), and the correlation matrix at the within- and between-person level are presented in Table 2 for all study variables. Next, to determine whether students' gender (boys versus girls), family structure (intact versus non-intact), and grade level (4th, 5th, or 6th grade) were associated with our study variables, we estimated a multilevel model with these demographic variables as between-person predictors and with our study variables as dependent variables. First, the results showed that gender was significantly associated with controlled motivation ($B = -0.18$, $p = .024$), engagement ($B = 0.17$, $p < .001$), and defiance ($B = -0.20$, $p < .001$). More specifically, boys reported more controlled

motivation, less engagement, and more oppositional defiance than girls. Second, we found a significant association between grade level and engagement ($B = -0.11$, $p = .029$), indicating that students in higher grades were less engaged than students in lower grades. Finally, family structure was significantly associated with oppositional defiance ($B = -0.13$, $p = .042$), compulsive compliance ($B = -0.25$, $p = .013$), and accommodation ($B = -0.27$, $p = .010$). More specifically, students whose parents were divorced reported less oppositional defiance, less compulsive compliance, and less accommodation. Based on these results, we decided to control for students' gender, grade level, and family structure in all further analyses.

Primary Analyses

Prior to addressing the role of coping, we estimated the main associations among perceived controlling teaching and the outcome variables (see Table 2 in appendix). This model had an excellent fit: CFI = 0.99, RMSEA = 0.021, SRMR = 0.00 at the within-person level, and SRMR = 0.034 at the between-person level. At the within-person level, the results first showed that perceived controlling teaching is related to motivational outcomes, that is, less autonomous motivation, more controlled motivation, and more amotivation. Second, perceived controlling teaching was unrelated to the behavioral outcome (engagement). At the between-person level, perceived controlling teaching was related to motivational outcomes, but only to more controlled motivation and more amotivation. Again, we did not find an association between perceived controlling teaching and engagement.

Aim 1: The Moderating Role of Coping. The model had a good fit with a CFI equal to 0.90, an RMSEA smaller than 0.04, and an SRMR smaller than 0.06 at both the between- and within-person level. In total, we found 4 significant interactions, three of which were situated at

the within-person level, and one of which was situated at the between-person level (see Figures 2-5 and Table 3).

At the within-person level, two out of the four moderation effects dealt with amotivation as an outcome and one dealt with engagement as an outcome. First, the results showed that the positive association between perceived controlling teaching and amotivation was stronger at higher levels of oppositional defiance (i.e., an exacerbating effect; see Figure 2), whereas this association became smaller when students reported higher scores on accommodation (i.e., a buffering effect; see Figure 3). Second, the results showed that the association between perceived controlling teaching and engagement changed from non-significant to negative when students reported lower scores on accommodation (see Figure 4).

At the between-person level, the moderation effect dealt with engagement as an outcome. Specifically, we found that the relation between perceived controlling teaching and engagement changed from non-significant to negative for students lower on compulsive compliance and from non-significant to positive for students higher on compulsive compliance (see Figure 5).

Aim 2: The Mediating Role of Coping. The mediation model had an excellent fit with a CFI equal to 0.98, an RMSEA equal to 0.02, and an SRMR smaller than 0.001 at the within-person level, and an SRMR equal to 0.05 at the between-person level. Results of the mediation model can be found in Table 4 and Figure 6. In total, we found that 10 out of 32 possible indirect effects were significant, two at the within-person level, and eight at the between-person level. Five significant indirect effects were found for oppositional defiance, three for compulsive compliance, and two for negotiation. First, at the within-person level, we found an indirect association between perceived controlling teaching and amotivation and engagement through oppositional defiance. Specifically, more perceived controlling teaching in a given week

(compared to their individual average) was related to more oppositional defiance that week (compared to their individual average), which, in turn related positively to amotivation and negatively to engagement. Second, at the between-person level, we found an indirect association between perceived controlling teaching and controlled motivation, amotivation, and engagement through oppositional defiance. More specifically, students who perceived more controlling teaching than others also displayed more oppositional defiance, which in turn related positively to amotivation, and negatively to controlled motivation and engagement. Next, we found an indirect association between perceived controlling teaching and autonomous motivation, controlled motivation, and amotivation through compulsive compliance. More specifically, students who perceived more controlling teaching also reported more compulsive compliance, which, in turn was positively related to controlled motivation and amotivation, and negatively related to autonomous motivation. Finally, we found an indirect relation between perceived controlling teaching and autonomous motivation and engagement through negotiation. Specifically, students who perceived more controlling teaching reported more negotiation, which, in turn, related positively autonomous motivation and engagement. Note that we also found a positive direct association between perceived controlling teaching and controlled motivation and amotivation at the within-person level. At the between-person level, there was a positive direct association with controlled motivation. To examine the stability of the findings, two sets of ancillary analyses were performed. In a first additional analysis, we have split the mediation model (including all coping responses and outcome variables) into 4 models (one per coping strategy) to test whether a model with only one mediating variable would give different results. Only one indirect association become non-significant when examining the coping strategies separately. Specifically, the indirect association between controlling teaching and

controlled motivation through oppositional defiance at the between-person level was borderline significant ($p = .058$) when looking at oppositional defiance separately. All other results did not change. In a second additional analysis, we added autoregressive associations for the outcome variables to conduct a more conservative test of the mediation analyses. As a result, four indirect associations at the between-person level were no longer significant. More specifically, the indirect association between controlling teaching and controlled motivation through oppositional defiance was no longer significant. The indirect associations between controlling teaching and autonomous motivation through compulsive compliance and through negotiation were no longer significant. Finally, the indirect association between controlling teaching and amotivation through compulsive compliance was no longer significant. The results of these additional analyses were also added in the appendix (see Table 3-4 and Figure 1).

Discussion

In recent years, research on the dark side of teaching has gained prominence, with several studies documenting the adverse consequences associated with controlling teaching (e.g., Haerens et al., 2015). In spite of this growing research interest, there remain two gaps in this body of literature. First, there is a dearth of studies examining the consequences of controlling teaching at both the between- and within-person level of analysis (see Bartholomew et al., 2018 for an exception). Second, most extant studies examined controlling teacher-student dynamics from a unidirectional perspective, thereby ignoring the active role students may play in these dynamics. By focusing on the role of coping, this study advances our understanding of the highly complex transactions between controlling teaching and student outcomes.

Educational Outcomes and Coping Responses Associated with Controlling Teaching at both the Inter- and Intra-individual Level

In line with Bartholomew et al. (2018) and Patall et al. (2018), the results showed that when students perceive their teacher as more controlling in a given week, compared to their own intra-individual average, **they simultaneously report lower levels of autonomous motivation and higher levels of controlled motivation and amotivation compared to their individual average.** All of these associations also occurred also at the between-person level of analysis, except for the negative association between controlling teaching and autonomous motivation. Also, there was no significant association between controlling teaching and engagement at either level of analysis, possibly because engagement is a more distal outcome. **In general, our results show that teachers are perceived to differ on a weekly basis in terms of the amount of control they exert on their students,** with controlling practices relating meaningfully to ups and downs in students' motivation. Overall, these findings are in line with theoretical predictions (Ryan & Deci, 2017) and highlight that a controlling teaching style is not entirely set in stone.

Next, controlling teaching was related to students' coping responses in theoretically predicted ways. As expected and in line with De Meyer et al. (2016a), Haerens et al. (2015), and studies in the context of parenting (e.g., Van Petegem et al., 2015), perceived controlling teaching was related to more oppositional defiance, both at the level of weekly variation and at the level of interindividual differences. Additionally, we found that controlling teaching is related positively to compulsive compliance, at both levels of analysis. **These results suggest that controlling teaching goes hand in hand with a climate in which children are alienated from their inner values and priorities, and feel pressured to either obey rigidly or to act-out and defy the authority altogether.** One possibility is that children combine both responses and switch back and forth between submissive conformity and rebellious disobedience. Perhaps initially children comply submissively as long as possible but at some point they resort to a sudden outburst of

defiance and rejection. Another possibility is that controlling teaching gives rise to oppositional defiance for some children, whereas elicits compulsive compliance for other children. Such a pattern of multifinality could be explained by time-invariant variables, such as children's temperament. For example, when confronted with a controlling teacher, children with a high behavioral activation system (BAS), which is characterized by approach behaviors and impulsiveness, may be more likely to react with oppositional defiance, whereas children with a high behavioral inhibition system (BIS), which is characterized by avoidance behaviors and anxiety, may be more likely to react with compulsive compliance (Gray, 1987).

Controlling teaching was related not only to non-autonomous coping strategies, but also to negotiation, albeit only at the between-person level. This association indicates that students who perceive their teacher as more controlling, are more likely to rely on negotiation. This finding was rather unexpected, because previous research in the context of parenting found a negative association between controlling parenting and adolescents' negotiation (Author, 2020). One possible explanation for this finding is that the majority of children participating in this study were raised in a generally autonomy-supportive home environment and have been taught to cope with threats to their autonomy in an autonomous manner (Van Petegem et al., 2017). When children from autonomy-supportive homes are confronted with a controlling teacher, they could then on average be inclined to react to such teaching with negotiation. A study by Van Petegem et al. (2017) indeed showed that adolescents' response to a specific controlling situation (as induced by a vignette) depended on the general parenting context in which the adolescent was raised. Adolescents who generally perceived more autonomy support at home responded with more negotiation to the (occasional) controlling situation. A second possible explanation is that controlling teaching does not elicit negotiation, but that negotiation elicits more controlling

teaching. Possibly, teachers do not always positively appraise students' negotiation and even respond negatively to it. This would be especially likely for teachers that are quite authoritarian in nature. Another possible explanation for this finding is that elementary school children are not yet able to accurately assess their negotiation behavior. Whereas children may perceive their 'negotiation' as a constructive dialogue with their teacher, their behavior may actually resemble oppositional defiance from the teacher's perspective. This possibility is in line with research showing that children's self-perceptions of ability gradually become more accurate (in terms of a stronger correlation between perceived competence and actual performance; Bouffard et al., 1998). To test the possibility that age affects the accurate appraisal of negotiation, we examined whether the association between perceived controlling teaching and negotiation depended on grade level (4th, 5th, or 6th grade; see Table 5 and Figure 2 in Appendix). We indeed found a significant interaction effect, where the association between controlling teaching and negotiation was weaker (and not significant) for children in 6th grade. The association was strongest for children in 4th grade. It should be noted that both of the above explanations are speculative at the time being. Future research that adopts an observational design or that takes the parental context into account could shed more light on the surprising positive association between controlling teaching and negotiation.

Finally, the results did not reveal any significant associations among controlling teaching and accommodation. As such, children do not typically respond to controlling teaching by accommodating the pressuring teacher demands. It is of course possible that some children, yet not all, do engage in accommodation in response to controlling teaching and an important question is whether such a coping strategy helps to buffer the negative effects of controlling

teaching. Accordingly, we now turn to a discussion of the precise role of accommodation and the other coping responses in associations **among** controlling teaching and the educational outcomes.

The Role of Students' Coping in Effects of Controlling Teaching

The main goal of this study was to examine the intervening role of the coping strategies when students were confronted with controlling teaching, with this intervening role taking the form of moderating and/or mediating effects. Consistent with previous theorizing and research on coping in general (Holmbeck, 1997; Jose & Huntsinger, 2005), we found that coping with controlling teaching can simultaneously play a moderating and a mediating role.

First, oppositional defiance was found to simultaneously exacerbate (i.e., moderate) and explain (i.e., mediate) associations **among** perceived controlling teaching and amotivation at the within-person level of analysis. An higher-than-average use of oppositional defiance thus amplified the strength of associations **among** controlling teaching and amotivation. Additionally, it explained this association. **In other words, when children perceived higher-than-average controlling teaching in a given week, they reported correspondingly higher-than-average oppositional defiance, which in turn was related to more amotivation in that week.** Because oppositional defiance plays a mediating role, the results suggest that oppositional defiance may contribute to the maintenance of a negative vicious cycle, with lack of motivation following from it eliciting teachers' further engagement in controlling practices (Sarrazo et al., 2006; Van den Berghe et al., 2016). Because oppositional defiance simultaneously plays a moderating role, it may at the same time worsen this cycle, resulting in an escalating deterioration of the teacher-student relationship.

At the between-person level, oppositional defiance exclusively played a mediating role. Students who generally use more oppositional defiance reported more amotivation, less

engagement, but also less controlled motivation (mediation). Surprisingly, we did not find any moderation effects of oppositional defiance at the between-person level. A closer look at the meaning of these two levels can shed light on these results. When students are confronted with temporary higher-than-average levels of controlling behavior, a corresponding higher use of oppositional defiance may lead to mutual trust issues, in which both the teacher and the student feel sold short and not understood. Indeed, studies have shown that defiant behavior relates negatively to students' trust in their teacher (Gregory & Ripski, 2008) and that teachers are more sensitive to defiant behavior when they are stressed (Kokkinos et al., 2005). This reciprocal negative appraisal of each other's behaviour and intentions might lead to a stand-off in which both sides fail to see their common goals. On the other hand, when students are confronted with a chronically controlling teacher, oppositional defiance might be at least somewhat protective because it helps students to distance themselves from the teacher's intrusive and domineering tactics. A recent qualitative study with 8-12 year old children showed that many children indeed perceived defiance as a legitimate way to protect themselves from teachers' unjustified control (Gingo, 2017).

With respect to compulsive compliance, we only found effects at the between-person level. More specifically, the significant interaction effect showed that, although the average association between perceived controlling teaching and engagement was non-significant, the association differed according to students' levels of compulsive compliance. When looking at students with higher levels of compulsive compliance, the association became more positive, whereas the association became negative for students low on compulsive compliance. **Although unexpected at first sight, this interaction effect suggests that students high on compliance are more resilient to controlling teaching practices. This should be nuanced, however, as compulsive**

compliance is only associated with high levels of class engagement in the face of controlling teacher practices, but not with high-quality motivation. Because compulsive compliance did not display any moderating effect at the level of within-person variation, it can be concluded that this strategy did not play a substantial moderating role in associations among controlling teaching and student outcomes.

The mediating role of compulsive compliance was more prominent, but only at the between-person level. Compulsive compliance partly explained the positive associations among perceived controlling teaching and controlled motivation and amotivation. Additionally, compulsive compliance also partly explained the negative associations among perceived controlling teaching and autonomous motivation. These results indicate that higher use of compulsive compliance goes hand in hand with more than usual (low quality) motivation and less than usual high quality motivation. These mediation effects are in line with the assumed dysfunctional role of compulsive compliance (Skinner et al., 2003).

Regarding negotiation, we found that it did not moderate any associations with perceived controlling teaching, neither at the within-person level, nor at the between-person level. The lack of a moderating role of negotiation could be due to the highly interactional nature of negotiation (Author, 2020). The role of negotiation probably depends strongly on how the teacher reacts to it. Future research would do well to examine how teachers perceive and respond to negotiation, in order to gain more insight into the boundary conditions of students' negotiation. Regarding the mediating role of negotiation, we found that perceived controlling teaching was associated indirectly with the two positive student outcomes through negotiation at the between-person level (i.e., autonomous motivation and engagement). More specifically, these effects showed that perceived controlling teaching elicits more negotiation, which in turn elicits more autonomous

motivation and engagement. When looking at the direct effects, perceived controlling teaching was related negatively to autonomous motivation, and unrelated to engagement. As such, these findings suggest that, instead of being unidirectionally and uniformly affected by controlling teaching, through negotiation some positive outcomes can be obtained, indicating that students are active agents in teacher-student dynamics. Yet, these positive indirect effects should also be interpreted cautiously, because the indirect associations through negotiation only accounted for 11 and 16% of the total effect of perceived controlling teaching on autonomous motivation, and engagement, respectively.

Regarding accommodation, we found a buffering role with respect to amotivation and engagement at the within-person level. As hypothesized, this effect indicates that if students, in a given week, rely more on accommodation when they perceive their teacher as being more controlling than usual, they suffer less from amotivation and low engagement. This is an interesting finding, because, when student accommodate, this seems to allow them to mobilize and energize themselves. That is, when being exposed to controlling events, quantitatively speaking they remain more motivated, as indexed by lower levels of amotivation and higher levels of engagement. This might be the case because accommodation helps them to identify how the request of the teacher may be well-intended and beneficial for their learning trajectory. It is important to note, however, that accommodation did not display a similar buffering effect on the quality of motivation (i.e., controlled or autonomous motivation). Hence, although accommodation appears to go hand in hand with a high quantity of motivation and engagement in the face of controlling teaching, it does not necessarily go hand in hand with a better quality of motivation. In other words, it is not because students manage to accommodate with a controlling event, that they feel good about it, or start to enjoy and value the lesson. As such, there are limits

to the extent to which accommodation can protect students against the problems associated with controlling teaching.

Moreover, accommodation failed to play a buffering role at the between-person level of analysis. One possible explanation for this lack of interaction effect at the between-person level is that when students are confronted with a controlling teacher on a more chronic basis, and they persistently engage in accommodation, they may get alienated from their own long term goals and perspective, as they constantly (and ineffectively) adjust their own preferences in the service of the teacher's demanding expectations. This reasoning is in line with the dual-process model of assimilative and accommodative coping, which suggests that accommodative coping processes (goal adjustment) should be complemented with assimilative coping processes (goal pursuit) in order to optimize coping effectiveness (Brandtstädter & Rothermund, 2002; Freund & Baltes, 2002). Based on these findings, it seems that accommodation is not inherently good or bad, but is time-dependent. Whereas accommodation might be somewhat effective on a momentary basis, in the longer run, it does not seem to be beneficial for students' quality of motivation.

Overall, the results of this paper are important both from a fundamental and an applied perspective. Indeed, the current study identifies students most at risk to suffer from a controlling teacher and offers suggestions for how students could optimally cope when confronted with a controlling teacher. Students' coping may be an innovative target for interventions aimed at teacher-student dynamics. Interventions addressing students' coping strategies may complement already existing interventions addressing teachers' styles. We argue that intervention-based research should aim for a 'both/and' approach, training both the teacher and the student, thereby setting in motion a reciprocal spiral of improving teacher-student interactions. In the end, the

combination of these two types of interventions may be greater than the sum of its parts and may reach the most beneficial effects.

Moreover, at a more theoretical level, the results from the current study have important implications for our understanding of teacher-student dynamics. Indeed, the current study shows that students actively shape their interactions with teachers, thereby impacting how they are affected by demotivating teaching styles (Soenens & Vansteenkiste, 2020). As past research has mostly examined teacher-student interactions from a unidirectional perspective, the current results suggests future research should take into account students' agency in teacher-student dynamics. Therefore, we argue that future research would do well to adopt a truly transactional perspective, according to which students can change the very nature of the relationship with their teachers.

Limitations and suggestions for future research

First, the reliability of some of our measures was rather modest, and unacceptably low in the case of disengagement, which caused us to drop this outcome. Importantly however, all questionnaires used in this study were short versions of well-validated measures. Moreover, confirmatory analyses including all study variables were performed for each wave. These models had a good fit and all factors loadings were significant at $p < .01$.

Second, teachers were aware that students filled out questionnaires on their perceived teaching style. This awareness could have impacted teachers' teaching behavior (both on purpose or unconsciously). Because we did not observe teachers' teaching style, we could not examine whether teachers changed their style. However, studies have shown that teaching do not change easily without intensive skill-based training (Reeve, 1998; Reeve et al., 2014). Moreover, to examine the consequences of students' coping with controlling teaching, the objective teaching

style may be of secondary importance. SDT argues that it is not the objective teaching style as such that will impact students' motivation or coping skills, but rather the functional significance, or psychological meaning provided by the student to the teacher's behaviour (Ryan & Deci, 2017). A controlling teacher can be perceived as very controlling or mildly controlling, depending on the recipient's characteristics (Vansteenkiste et al., 2020). For this reason, we argue that a possible change in teaching style, due to awareness of the objective of the study, may not have a substantial impact on our results (which are based on students' perceptions)..

Third, although need frustration has been proposed as a mediator of associations **among** controlling teaching and adolescents' outcomes (e.g., amotivation, engagement and school ill-being; Bartholomew et al., 2018; Haerens et al, 2015), the current study did not take into account need frustration. Similarly, whereas motivation has been proposed as an additional mediator (e.g., Vansteenkiste et al., 2005), the current study examined motivation as an outcome variable rather than as an intervening variable. However, future research could examine the role of coping in associations **among** controlling teaching and need frustration and motivation as a more proximal process variable. The moderating role of coping in these associations could be examined using a conditional process model, thereby examining whether coping moderates the indirect association between controlling teaching and students' outcomes through need frustration and motivation.

Fourth, we relied on students' self-reports only to measure all variables. This method could have inflated associations **among** above mentioned constructs due to shared method variance. Future research could include more objective measures, for example by observing children's coping or engagement.

Fifth, because this was one of the first studies to examine both the moderating and the mediating role of students' coping responses to controlling teaching, a large amount of associations were estimated. It will be important to replicate these associations in future research. The effect sizes found in the current study can then be used to perform power analysis.

Sixth, the coping items may be considered double-barrelled items, as they implicitly also ask about students' experiences of controlling teaching. These items may be difficult to answer for students who perceived controlling teaching to be very low. In an additional set of analyses, we checked whether this difficulty impacted our results by only including participants who had a score higher than one on perceived controlling teaching for at least three of the four measured waves. This led to an exclusion of 125 students. Overall, the results did not change, except for one indirect effect and one interaction effect. The loss of these effects may be due to the complexity of the coping items (because now, all participants reported, at least to some extent, experiences of controlling teaching), but it may also be due to a loss of power.

Finally, although the results suggest that teacher-student dynamics are highly transactional in nature, the present study was not able to directly test bidirectionality or direction of effects. Therefore, the results should be interpreted with caution. For instance, whereas the results of the paper suggest that controlling teaching elicits more oppositional defiance, which would in turn, elicit more amotivation, it is equally possibly that highly amotivated students are more likely to defy the teacher, which could in turn elicit more controlling teaching practices. Future research could examine how the results might differ when coping is assessed five minutes after the assessment of controlling teaching, and the motivational outcomes five minutes after the assessment of coping. Vice versa, future (preferably longitudinal) research could examine the

effect of the different coping responses on the teacher's teaching style, as measured by observations or teachers' self-reports.

Conclusion

This study showed that coping responses can - to some extent - alter the degree to which a controlling teaching style relates to students' maladjustment. **However, for each of the coping responses, we found a different pattern of moderation at the different levels of analysis, showing that the moderating role of coping responses depends on whether differences within students or differences between students are examined.** With respect to the mediating role of coping, findings support the idea of a bright and a dark pathway (Haerens et al., 2015), especially at the between-person level, with oppositional defiance and compulsive compliance being highly related to maladaptive student outcomes, and with negotiation being especially related to adaptive student outcomes. Overall, the results emphasize students' agency in teachers-student relationships, which is in line with SDT (Reeve, 2013) and the classroom ecology paradigm (Doyle, 1977; Hastie & Siedentop, 2006). Therefore, we argue that students' active role should be considered in future research and in future intervention efforts in order to capture the complexity of teacher-student dynamics.

Ethical approval: All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent: Informed consent was obtained from all individual participants included in the study.

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